

Improving PII Detection with DeBERTaV3 + DoRA

Fine-Tuning on Synthetic Data Targeting NER Failure Modes

CS 6140 Machine Learning — Final Project

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Date: December 2025

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Abstract

Personally Identifiable Information (PII) detection is critical for data privacy compliance, yet existing open-source models exhibit significant performance gaps on real-world data. Singh & Narayanan (2025) demonstrated that popular models like Piirinha and StarpII fail to detect PII in 28% of cases and misclassify entities 67% of the time. We address these gaps by fine-tuning DeBERTaV3-large using Weight-Decomposed Low-Rank Adaptation (DoRA) on a combined dataset of 156K examples, including 6,002 synthetically generated samples specifically targeting six NER failure mode dimensions. Our model achieves an F1 score of 0.908, outperforming the best existing open-source model (Piirinha, F1=0.244) by 272% while supporting 505 entity labels across 10 regional PII formats.

1. Introduction & Problem Statement

Privacy masking is a fundamental requirement for organizations handling sensitive data under regulations such as GDPR and CCPA. Named Entity Recognition (NER) approaches are widely used for PII detection, but they face significant challenges including content ambiguity, regional format variations, and adversarial inputs. Despite models like Piiranhya and Starpii having over 800,000 combined downloads on HuggingFace, recent research has exposed critical limitations in their real-world effectiveness.

Motivation: Singh & Narayanan (2025) evaluated three prominent PII masking models across 17K test samples and found: (1) 28% complete non-identification of any PII, (2) 67% misclassification of detected entities, and (3) particularly poor performance on non-US regional formats. These failures stem from sparse training data, lack of regional diversity, and insufficient coverage of emerging PII types like cryptocurrency wallets and UPI identifiers.

Objective: We aim to build a more robust PII detection model by: (1) combining the ai4privacy English dataset with synthetically generated challenging examples, (2) using DoRA fine-tuning for parameter-efficient adaptation, and (3) specifically targeting the six NER failure dimensions identified in prior work.

2. Data Collection & Preparation

Our training dataset combines two sources into 156,695 total examples:

Data Source	Train	Test	Description
ai4privacy (English subset)	120,533	30,160	Existing open-source PII dataset
Synthetic (Grok + GPT-5.1)	4,801	1,201	Generated targeting NER failure modes
Total	125,334	31,361	—

Table 1: Dataset composition

2.1 Synthetic Data Generation

We generated synthetic samples using Grok (xAI) with GPT-5.1 validation, targeting six NER failure dimensions from Singh & Narayanan's framework:

Dimension	Count (After Validation)	Description
Basic	989	Standard, well-formatted entities
Contextual	801	Ambiguous entities requiring context
Noisy	876	Typos, informal text, OCR-like errors
Evolving	754	Modern PII: crypto wallets, UPI, TOTP
Multilingual	917	International formats in English text
Adversarial	464	Intentionally deceptive patterns

Table 2: Synthetic data feature dimensions

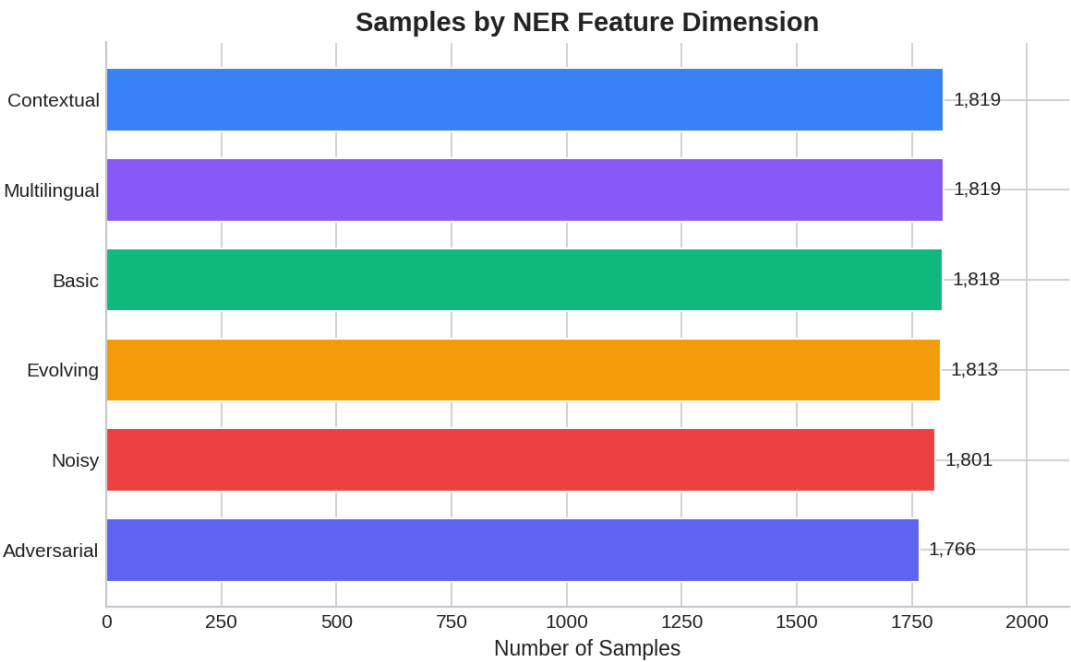


Figure 1: Synthetic Data Spread before validation using GPT 5.1

Generation Pipeline: Of 10,944 generation attempts, Grok successfully produced 10,836 samples (99.0%). After GPT-5.1 validation filtering for coherence, boundary accuracy, and label correctness, 6,002 samples (55.4%) were retained. Entity boundary accuracy was 96.6%, while label accuracy was 49.3% — the latter primarily due to schema mismatches (e.g., "PERSON" vs "NAME") resolved through post-processing.

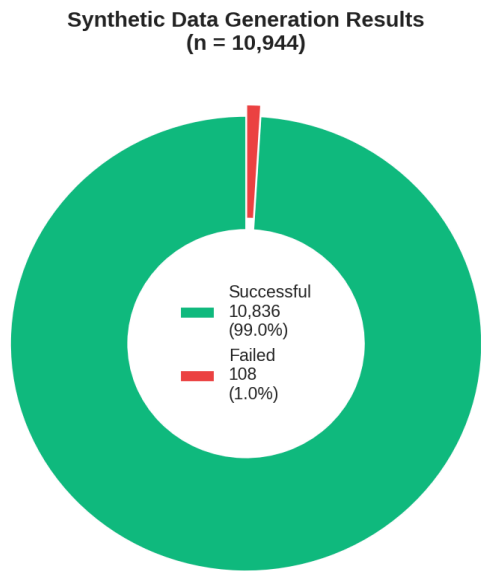
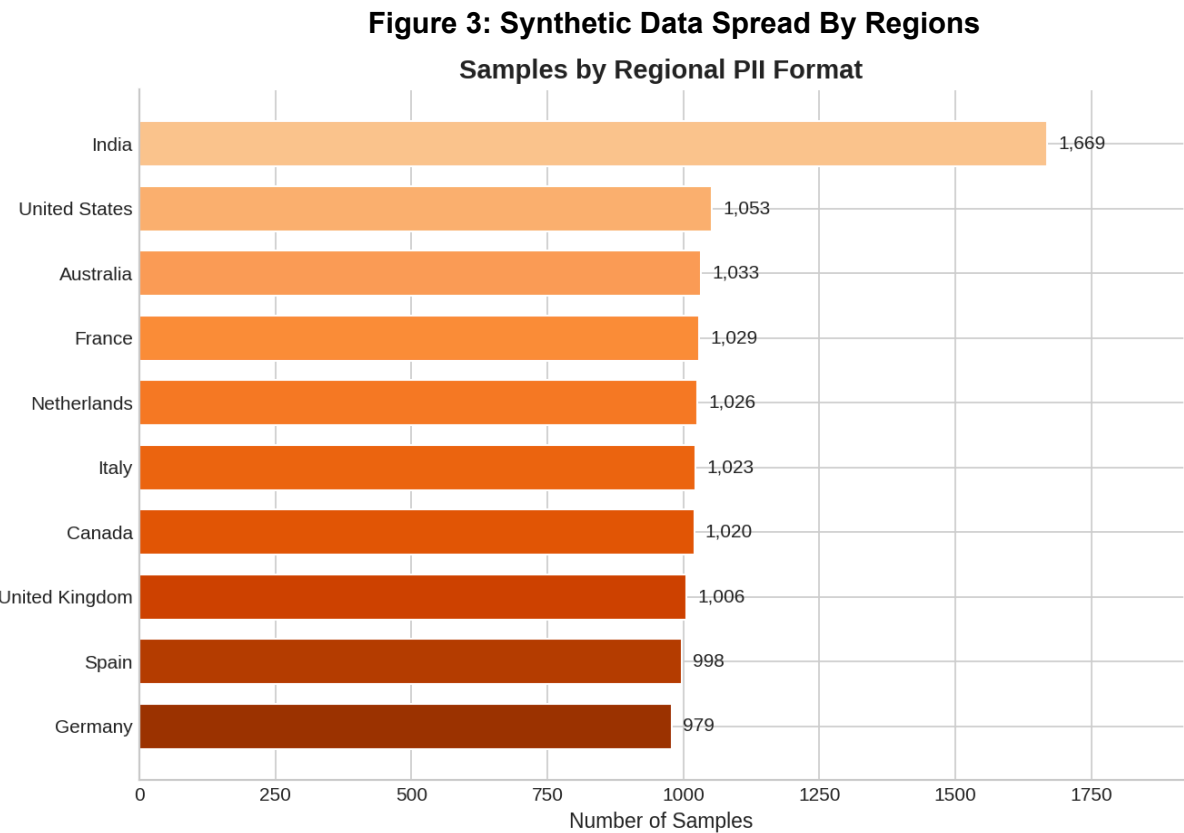


Figure 2: Synthetic Data Generation Results

Regional Coverage: Samples cover 10 locales (India, US, UK, Canada, Australia, Germany, France, Spain, Italy, Netherlands) with 16 PII types from Singh & Narayanan's taxonomy, plus emerging types like cryptocurrency addresses and UPI IDs.



3. Methodology

3.1 Model Architecture

We fine-tune **microsoft/deberta-v3-large** (304M parameters) for token classification using **Weight-Decomposed Low-Rank Adaptation (DoRA)**. DoRA decomposes pretrained weights into magnitude and direction components, applying low-rank updates only to the directional component. This approach has shown superior performance over standard LoRA for complex NLP tasks while maintaining parameter efficiency.

3.2 Model Configuration

Hyperparameter	Value
Base Model	microsoft/deberta-v3-large
Learning Rate	2e-5
Epochs	5 (with early stopping)
Batch Size	32
DoRA Rank (r)	16
DoRA Alpha	32
DoRA Dropout	0.05
Target Modules	query_proj, key_proj, value_proj, dense
Trainable Parameters	7.82M (2.6% of total)
Hardware	NVIDIA H100 (Modal)

Table 3: Training hyperparameters

Preventing Overfitting: We employed early stopping with patience=3 based on validation F1 score, DoRA regularization (dropout=0.05), and stratified train/test splits. Training completed at 19,000 steps with final training loss of 0.271.

4. Results

4.1 Model Comparison

We evaluated our model against four existing PII detection models on the held-out test set (31,361 samples):

Rank	Model	Precision	Recall	F1-Score	Label Count
1	DeBERTa-v3-large + DoRA (Ours)	0.898	0.918	0.908	505
2	mDeBERTa-v3-base (ai4privacy)	0.370	0.369	0.370	111
3	DistilBERT (ai4privacy)	0.342	0.368	0.355	116
4	Piirinha v1	0.600	0.153	0.244	18
5	DeBERTa PII (lakshyakh93)	0.100	0.164	0.125	116

Table 4: Model performance comparison on test set

Key Findings: Our model achieves an F1 score of **0.908**, representing a **272% improvement** over Piirinha (0.244), the best-performing existing model. Notably, Piirinha has high precision (0.600) but very low recall (0.153), meaning it misses the majority of PII entities — the exact failure mode identified by Singh & Narayanan. Our model maintains both high precision (0.898) and high recall (0.918), with overall accuracy of 97.9% and ROC AUC of 0.9998.

Label Coverage: Our model supports 505 entity labels compared to 18 for Piirinha, enabling fine-grained PII classification across names (first/middle/last), addresses (street/city/state/zip), financial identifiers (credit cards, bank accounts, tax IDs), and emerging PII types.

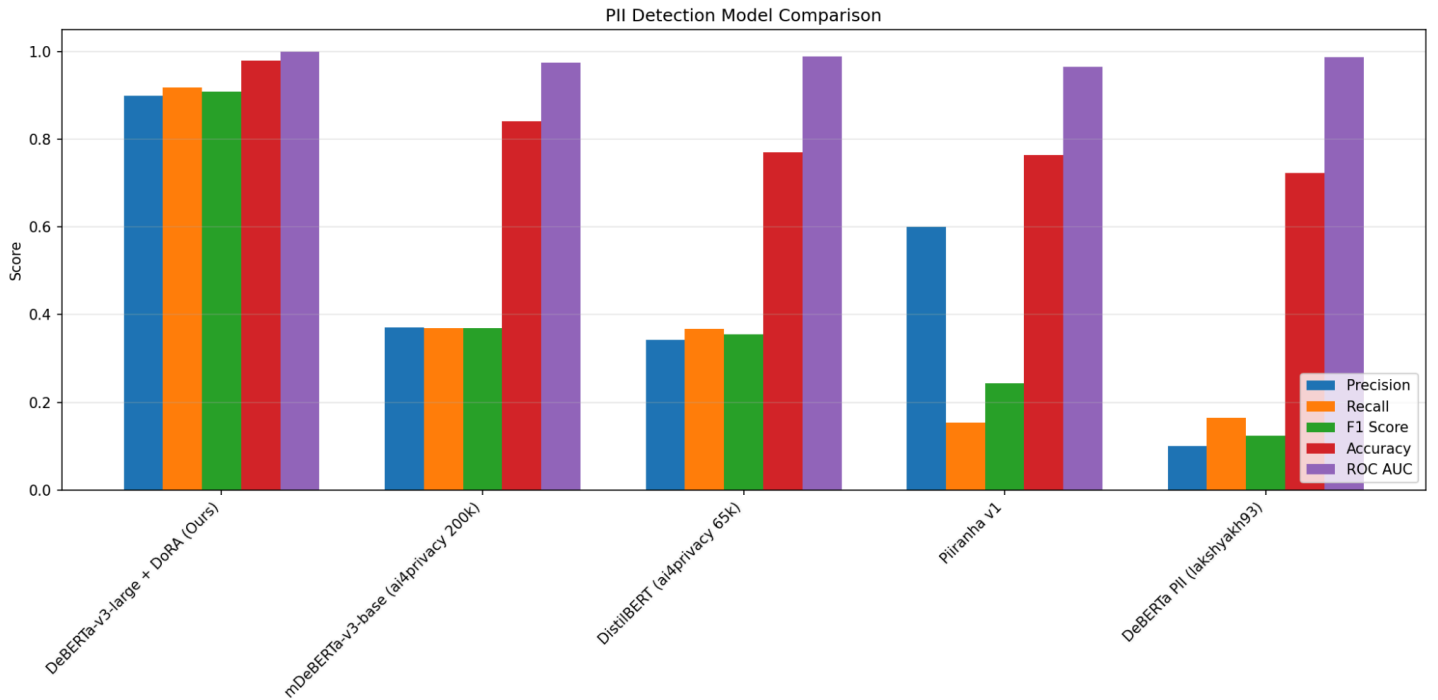


Figure 4: Model Comparison

5. Discussion

5.1 Why Our Approach Works

Three factors contribute to our model's strong performance: (1) **Targeted synthetic data** — by generating samples specifically for NER failure modes, we provide training signal for edge cases that existing datasets lack; (2) **DoRA fine-tuning** — weight decomposition enables efficient adaptation while preserving the pretrained model's linguistic knowledge; (3) **Larger base model** — DeBERTaV3-large has superior contextual understanding compared to DistilBERT or base-sized models.

5.2 Limitations

Synthetic data bias: Our multilingual dimension inadvertently included some non-English text that should have been filtered. **Label fragmentation:** The 505-label schema may be overly granular for some applications. **Computational cost:** DeBERTaV3-large requires

significant GPU memory for inference compared to DistilBERT. **Evaluation scope:** We evaluated on held-out test data from the same distribution; performance on truly out-of-distribution data (e.g., new PII formats) is untested.

5.3 Future Work

Potential improvements include: (1) Label consolidation to reduce the 505 labels to a more practical taxonomy, (2) Additional adversarial robustness testing, (3) Deployment optimization via quantization or distillation, and (4) Extending to multilingual text (not just international PII formats in English).

6. Conclusion

We presented a PII detection model that substantially outperforms existing open-source alternatives. By combining the ai4privacy dataset with synthetically generated samples targeting known NER failure modes, and using DoRA fine-tuning on DeBERTaV3-large, we achieved an F1 score of 0.908 — a 272% improvement over the previous best model. Our approach demonstrates that targeted synthetic data generation, guided by systematic analysis of model failures, can effectively address gaps in NER-based PII detection.

Resources

Model: huggingface.co/Ari-S-123/deberta-v3-large-pii-consolidated

Dataset: huggingface.co/datasets/Ari-S-123/pii-detection-english-consolidated

Code: github.com/Ari-S-123/pii-masking

References

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- [4] ai4privacy (2023). Open PII Masking Dataset. HuggingFace Datasets.